MASTER OF SCIENCE IN ENGINEERING

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Advanced Statistical Data Analysis
Worksheet Week 5

Exercise 1: Exponential Family. Let Y be a random variable with probability function given by

$$P\langle Y=y\rangle = \frac{1}{y!}\cdot\lambda^y\cdot e^{-\lambda} \quad \text{for } y=0,\,1,\,2,\,\ldots\,;$$

i.e., Y is Poisson distributed.

a) Show that the distribution of Y is a member of the exponential family by identifying the functions $b\langle\cdot\rangle$, $c\langle\cdot\rangle$, and $d\langle\cdot\rangle$.

b) Find the mean, variance, variance function $V\langle\cdot\rangle$ and canonical link for the Poisson distributions using the functions $b\langle\cdot\rangle$ and $c\langle\cdot\rangle$.

Exercise 2: The dial-a-ride data given in the file `Dial-a-ride.dat` are described in 4.2.o of the Lecture Notes.

a) Give a graphical and a numerical summary of the “raw” data.

b) Fit an ordinary linear regression model with Gaussian distributed responses using the original data. Inspect the residuals.

c) Improve the linear regression model by applying Tukey’s First-Aid transformations, then using the results of additive model fitting, and finally by applying robust fitting methods.

d) In class, a different model was proposed. Fit this model.

Exercise 3: The data given in the file `bacteria.dat` represents the number of surviving bacteria (in hundreds) as estimated by plate counts in an experiment with marine bacterium following exposure to 200 kilovolt X-rays for periods ranging from $t = 1$ to 15 intervals of 6 minutes. In the file, you find the number of surviving bacteria (`N` – in hundreds) and the time `Time` in units of 6 minutes:

t	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n_t	355	211	197	166	142	106	104	60	56	38	36	32	21	19	15

The experiment was carried out to test the single hit hypothesis of X-ray action under constant field of radiation. According to this theory, there is a single vital center in each bacterium, and this must be hit by a ray before the bacterium is inactivated or killed. The particular bacterium studied does not form clumps or chains, so the number of bacterium can be estimated directly from the plate counts. If the theory is applicable, the expected number of survivors should decay exponentially with time:

$$E\langle N_t\rangle = n_0 e^{\beta t}, \quad t > 0,$$

where n_0 and β are parameters. The parameters n_0 and β have simple physical interpretation; n_0 is the number of bacteria at the start of the experiment, while β is the destruction (decay) rate.

a) Formulate the "single-hit hypothesis" in the framework of generalized linear models (response, distribution, explanatory variables, link function).

b) Estimate the parameters of the model and discuss briefly the result (which parameters are significant and what is the interpretation of the quantitative values of the estimates?).

c) Calculate an approximate 95% confidence interval for β using the asymptotic normality of a maximum likelihood estimator - do it by hand (or with the R function `confint.default()`).

   Use the R function `confint()`. Do you get identical results?

Exercise 4: The transaction data given in the file `transactions.dat` are described in 4.2.s of the lecture notes.

a) Give a summary of the "raw" data.

b) Fit an ordinary linear model with normally distributed responses. Inspect the residuals. Why is this model not appropriate (in terms of a residual analysis)?

c) As described in 4.2.t of the lecture notes another model is suggested. Give the distribution of the response and the link function. Fit this model.

Exercise 5: Nambeware Polishing Times . Nambe Mills (http://www.nambe.com/) manufactures a line of tableware made from sand casting a special alloy of several metals. The polishing times for the products are thought to be related to the size of the item, as indicated by the diameter. After casting, the pieces go through a series of shaping, grinding, buffing, and polishing steps. In 1989 the company began a program to rationalize its production schedule of some 100 items in its tableware line. The total grinding and polishing times listed here were a major output of this program.

A dataset with 59 observations on the following 4 variables are given in the file `nambeware.txt`:

`Type`	the type of product; a factor with levels `Bowl`, `CassDish`, `Dish`, `Plate` and `Tray`
`Diam`	the diameter of the product in inches; a numeric vector
`Time`	the total grinding and polishing time in minutes; a numeric vector
`Price`	the price in us dollars; a numeric vector

a) Fit a gamma regression model to the response `Time` using a log link and a linear combination of the expalatory variables `Diam` and `Type` as linear predictor. What are the estimated coefficients?

b) How do you interpret the fitted model in part (a) with respect to the expected response?

   How would you interpret a gamma regression model with linear predictor (in R notation – see also 2.F.f in the Lecture Notes)

   `Time ~ Diam * Type` ?